

SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #25 – Smart Cities, Transport & Logistics

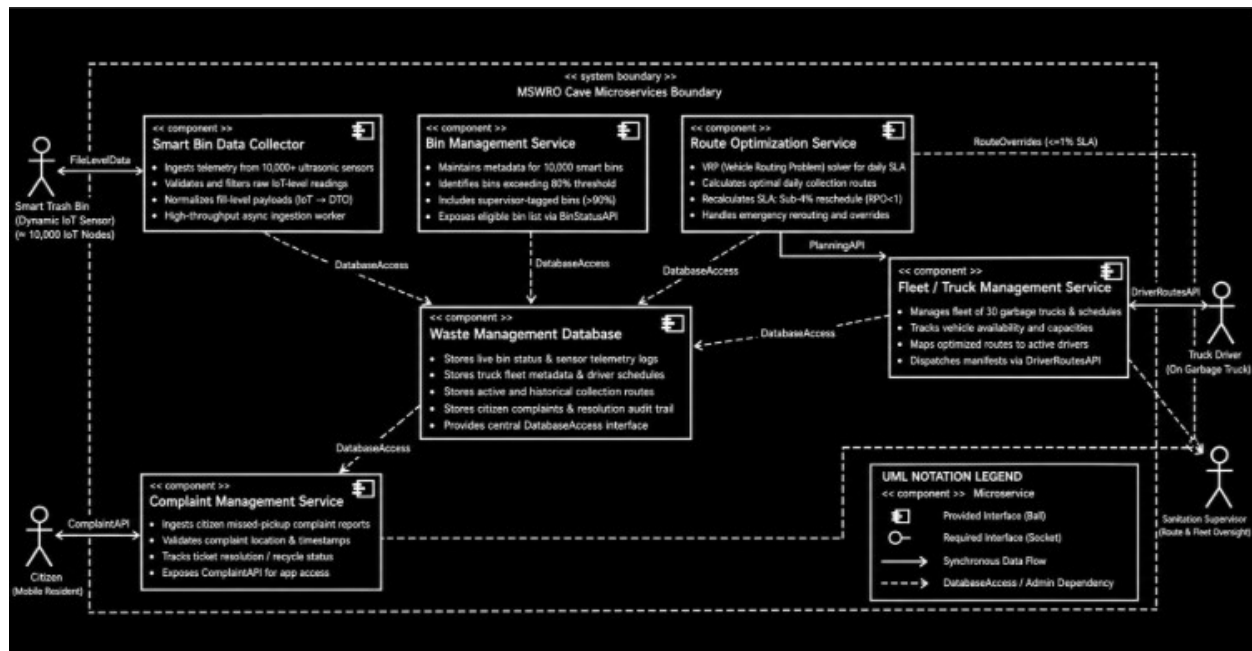
Municipal Solid Waste Route Optimizer

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1. Component Diagram



2. Architectural Justification

Architecture Selection

We chose Microservices Architecture for the Municipal Solid Waste Route Optimizer System.

Architectural Choice

The municipal solid waste route optimization pipeline is decomposed into independently deployable services aligned with the system's core capabilities — Smart Bin Data Collector, Bin Management Service, Route Optimization Service, Fleet / Truck Management Service, Complaint Management Service, and Waste Management Database — each communicating through well-defined interfaces (FillLevelData, BinStatusAPI, RoutePlanningAPI, DriverRouteAPI, ComplaintAPI, DatabaseAccess) rather than through a single monolithic codebase or a single centralized server.

Reason 1 — Independent Scaling Under High-Volume Telemetry & Solver Load (NFR-001 & Scale)

The Municipal Solid Waste Route Optimizer manages 10,000 smart trash bins equipped with ultrasonic sensors continuously transmitting fill-level telemetry, while simultaneously calculating optimized daily routes for 50 garbage trucks. NFR-001 requires that route recalculations for the fleet of 50 trucks across 10,000 bins execute in under 45 seconds. Because telemetry ingestion and route optimization are decoupled into separate services in a microservices architecture, the high-frequency IoT ingestion workload (Smart Bin Data Collector) can be scaled horizontally on asynchronous worker nodes without consuming the CPU and memory resources required by the computational VRP graph solver (Route Optimization Service). A layered or client-server design would force the entire application tier or single server to compete for shared thread pools and memory, causing continuous sensor ingestion to starve the optimizer and violating the sub-45-second route recalculation SLA.

Reason 2 — Independently Evolving Business Capabilities & Operational Fault Isolation

The waste management portal serves distinct stakeholders with vastly different operational rhythms: IoT sensors reporting raw fill levels every 15 minutes, Citizens submitting missed-pickup complaints, Sanitation Supervisors triggering manual route recalculations or overrides, and Truck Drivers receiving turn-by-turn route manifests. Each stakeholder interaction maps cleanly to a dedicated service (Smart Bin Data Collector, Complaint Management Service, Bin Management Service, Fleet / Truck Management Service) that can be developed, scaled, and deployed independently. This modular decomposition guarantees robust fault isolation: if a surge in citizen complaints causes a temporary slowdown or outage in the Complaint Management Service, the critical telemetry pipeline (Smart Bin Data Collector -> Bin Management Service) and active truck routing pipeline (Fleet / Truck Management Service -> Truck Driver) remain 100% operational, preventing citywide sanitation disruptions.

Security Advantage

Isolating persistent system data behind a dedicated Waste Management Database component, accessed strictly through a controlled internal interface (DatabaseAccess) rather than direct storage access from

external web endpoints, significantly narrows the attack surface. Public-facing APIs (such as ComplaintAPI used by Citizens) are decoupled from internal microservices, preventing unauthorized exposure of sensor hardware metadata, driver location tracking, or database credentials. Each service operates under the principle of least privilege, ensuring that even if a public-facing service is compromised, malicious actors cannot perform direct administrative overrides or access sensitive fleet data.

Performance Benefit

Decoupling the Route Optimization Service from the Smart Bin Data Collector and Complaint Management Service ensures that the most compute-intensive operation — running the Vehicle Routing Problem (VRP) solver to calculate or recalculate routes for 50 trucks across 10,000 bins under 45 seconds — is never blocked by slower, I/O-bound operations such as raw sensor data ingestion, photo evidence validation in citizen complaints, or database reporting writes. Intermediate bin state is evaluated and cached by the Bin Management Service and served to the solver in-memory via BinStatusAPI. This architectural separation provides dedicated CPU execution for graph optimization algorithms, guaranteeing that peak sensor ingestion volumes do not introduce latency jitter into emergency route recalculations.